Estimating Musical Surprisal from Audio in Autoregressive Diffusion Model Noise Spaces

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[2] Institute of Computational Perception

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JKU Institute of Computational Perception

ABOUT US RESEARCH TEACHING

- Music Information Research (MIR)
- Sound and Music Computing (SMC)
- Acoustic Scene & Event Recognition (DCASE)
- Recommender Systems (RS)

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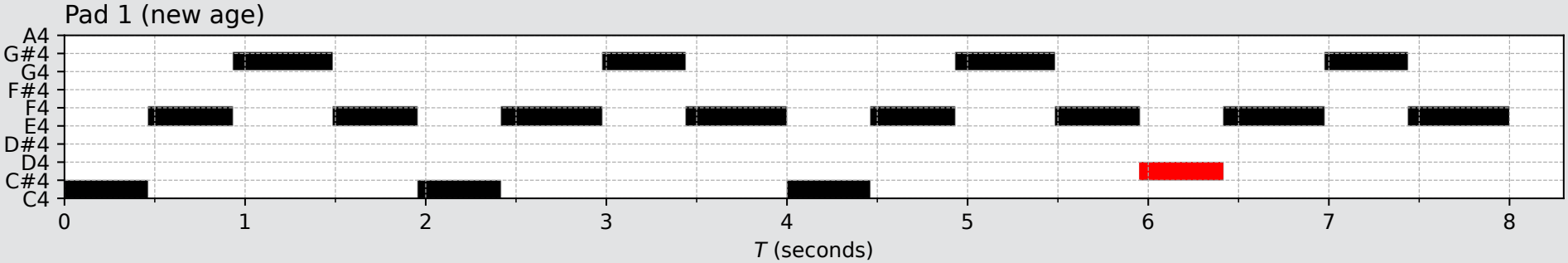
Pitch surprisal



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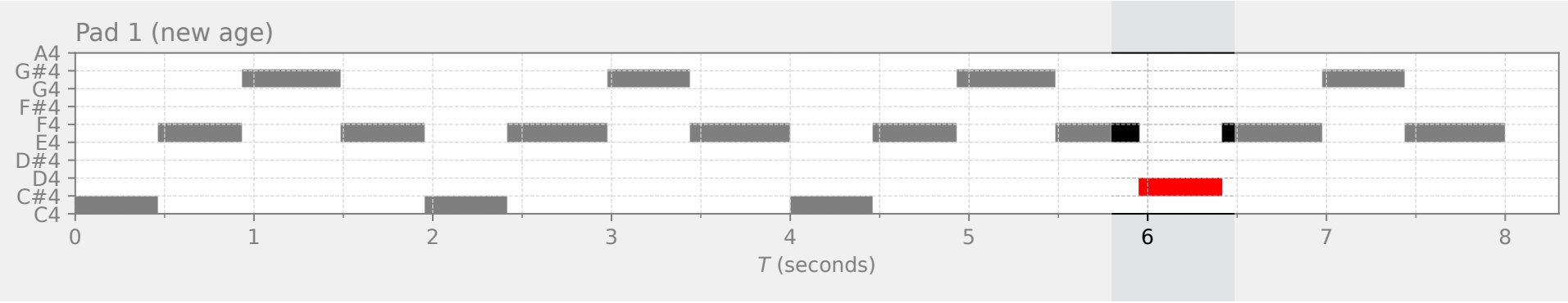

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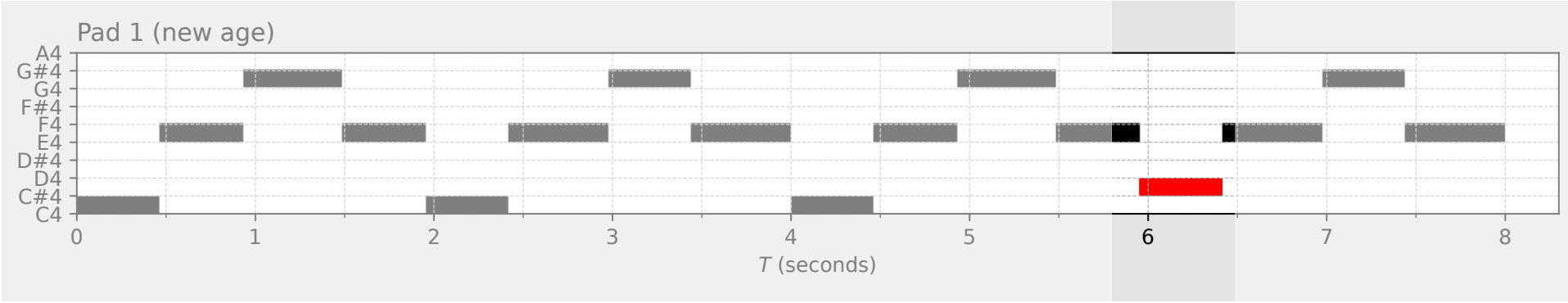

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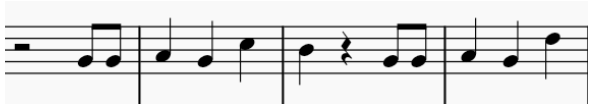
Model listening process by autoregressive prediction

Listener



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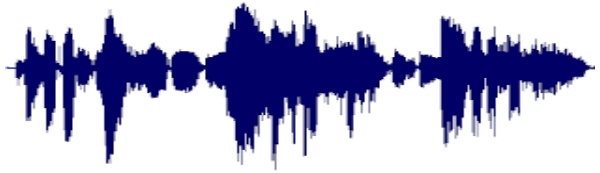

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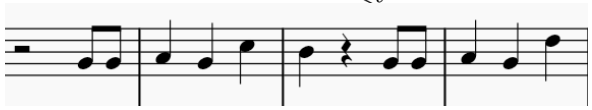
Model listening process by autoregressive prediction

$$p(x_t | x_{<t}) = g_\theta(x_{<t})$$

$x_{<t}$



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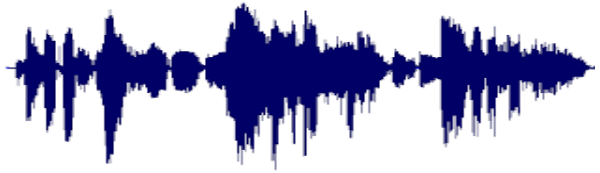

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Analysis of listening process

Given an autoregressive model, we can ask questions about the musical piece using Information Theory, following Meyer 1957[1] and Conklin and Wittens 1995[2].

- How surprising is a musical event to the listener, i.e. how high is the information content of the event

$$IC(x_t|x_{<t}) = -\log p(x_t|x_{<t})$$

- What is the overall uncertainty of the prediction, i.e. how high is the entropy:

$$H(x_t|x_{<t}) = -\sum_x p(x_t) \log p(x_t|x_{<t})$$

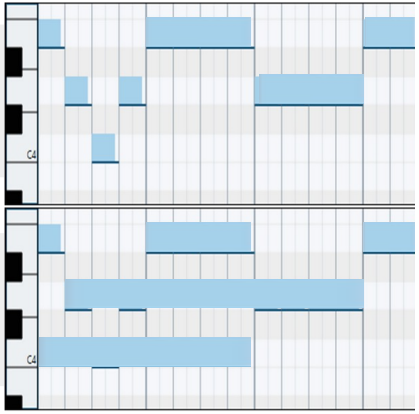
[1]Meyer, L. B. Meaning in music and information theory. *The Journal of Aesthetics and Art Criticism*, *15*(4), 412-424.,1957
[2]Conklin, D., & Witten, I. H. Multiple viewpoint systems for music prediction. *Journal of New Music Research, 24*(1), 51-73. 1995

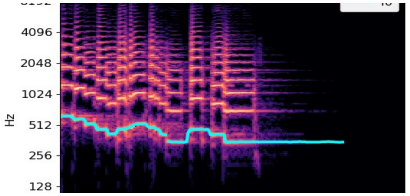
Surprise estimation with music features

1 Monophonic Symbolic Music
- IDyOM[1]

2 Polyphonic Symbolic Music
- Instantaneous Information Content[2]

3 Audio features
- D-REX[3]





NLP-like

CV-like

[1] Conklin, D., & Witten, I. H., Multiple viewpoint systems for music prediction. *Journal of New Music Research, 24*(1), 51-73., 1995
[2] Bjare, MR., et al. Controlling surprisal in music generation via information content curve matching, ISMIR, 2024.
[3] Benjamin Skerritt-Davis and Mounya Elhilali. Detecting change in stochastic sound sequences. PLoS Comput. Biol., 14(5), 2018.

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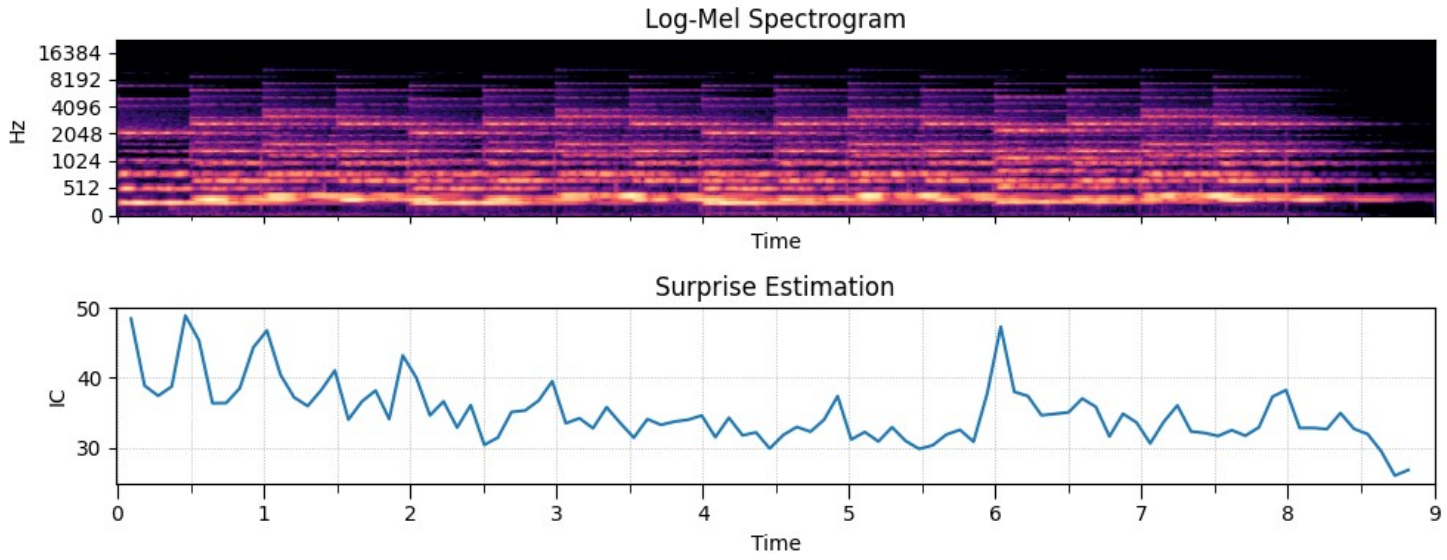
Challenges music feature-based surprise estimations

- What music features to select (pitch, rhythm, dynamics)?
- How to calculate/extract music features relevant to surprisal?
- How to combine multiple possibly highly correlated features?
- How to align surprise estimations with neural measurements from listener studies?

Challenges music feature-based surprise estimations

We propose circumventing these challenges by estimating surprise directly in audio



Measuring IC in Audio

Practical considerations:

Symbolic

MAESTRO (solo piano) dataset: ~5200 notes per piece

Audio

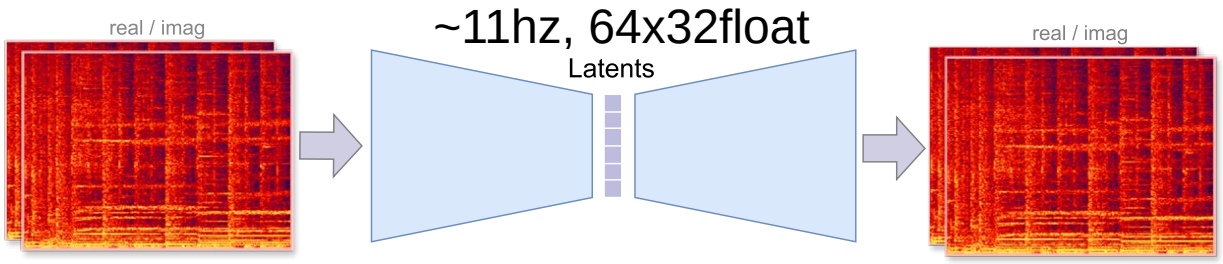
44.1kHz or 48kHz

It is **computational infeasible** to model long term effects in raw audio

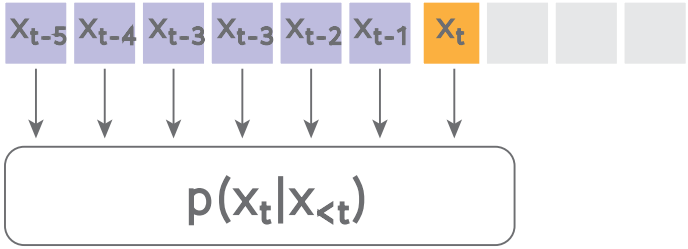
Measuring IC in Audio

2-stage approach:

1) Compress (wave) audio sequence using an **INVERTIBLE** transformation



2) Autoregressive next-element prediction



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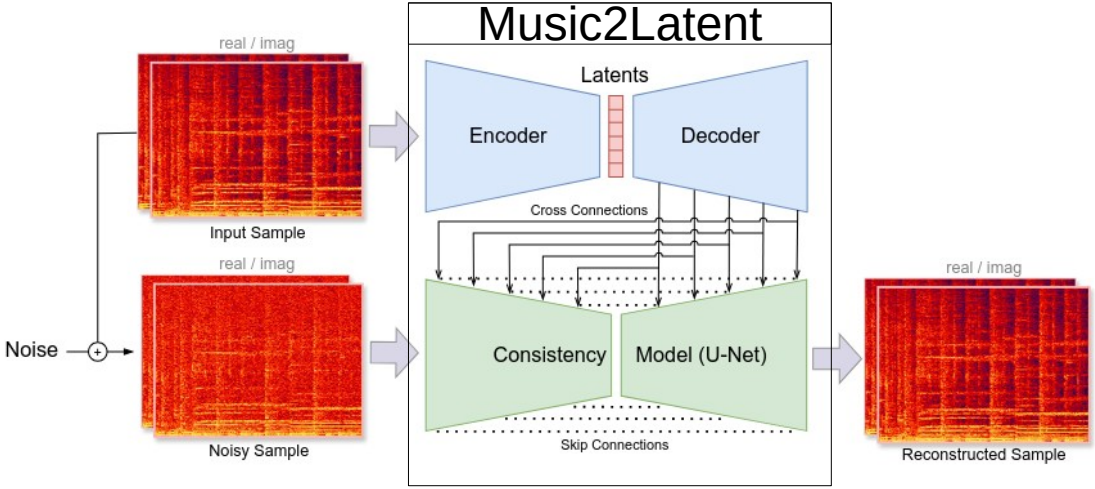
1. Stage – Music2Latent encoding

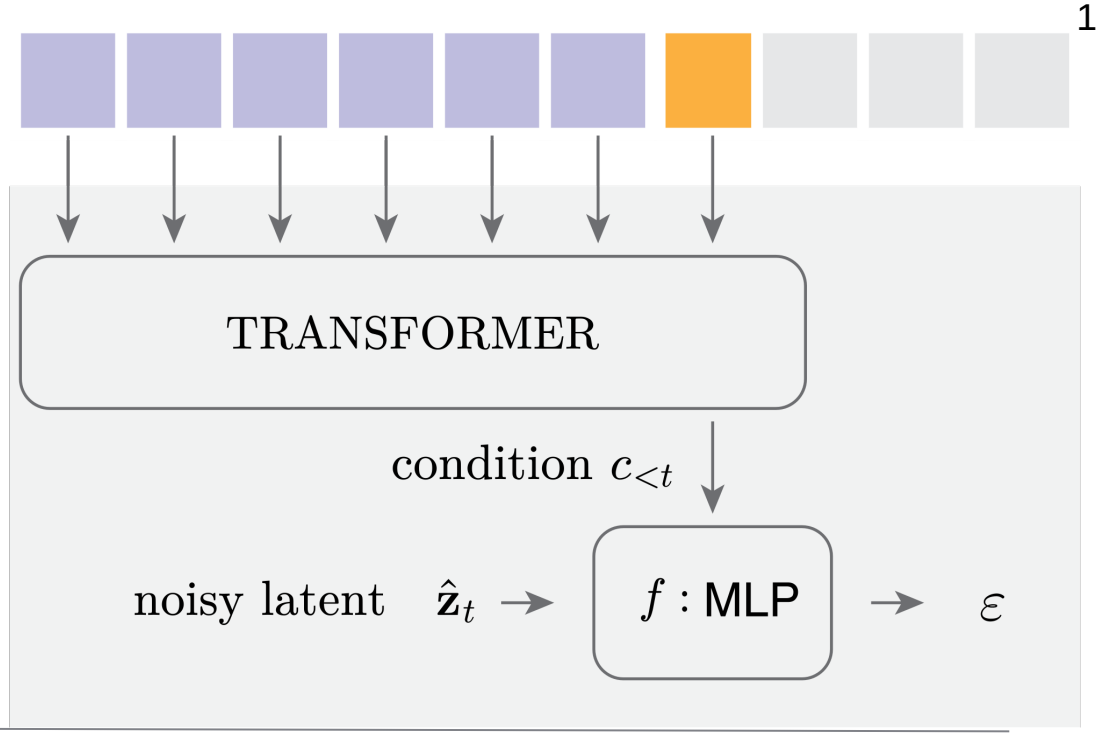
- An **autoencoder** – an artificial neural network trained to compress and reconstruct its input.
- Compresses audio of 44.1/48kHz to a **continuous** sequence of **64-channels** at **11Hz**
- Highly **entangled dimensions** – a single channel encodes many musical features
- A python package – pip install music2latent (https://github.com/SonyCSLParis/music2latent)



[1] Marco Pasini, et al. Music2latent: Consistency autoencoders for latent audio compression. In ISMIR, 2024.

2. Stage - Diffusion Model Audio Surprisal Estimation

- As the representations are continuous, we estimate **densities**, not **probability mass functions**
- Summarize the (musical) context of past observations as a vector $c_{<t}$
- Use the context $c_{<t}$ to condition a small rectified flow network *f* that learns to measure the liklihood of next-element z_t.



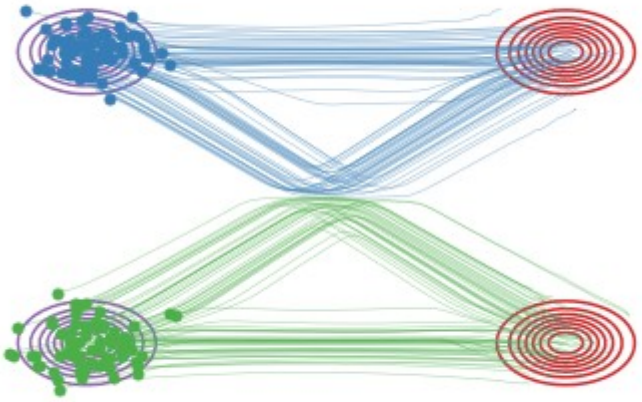
[1] Li, Tianhong et al., Autoregressive image generation without vector quantization. In NeurIPS, 2024.

Rectified flow[1,2]

- Transform points z1 of any distribution to points z0 of another distribution
- Learns a flow ODE between the distributions:

$$\mathrm{d}Z_t = f(Z_t, t)\mathrm{d}t$$

  , where *f* is a neural network.
- Flow points by taking small steps in the direction of *f*.



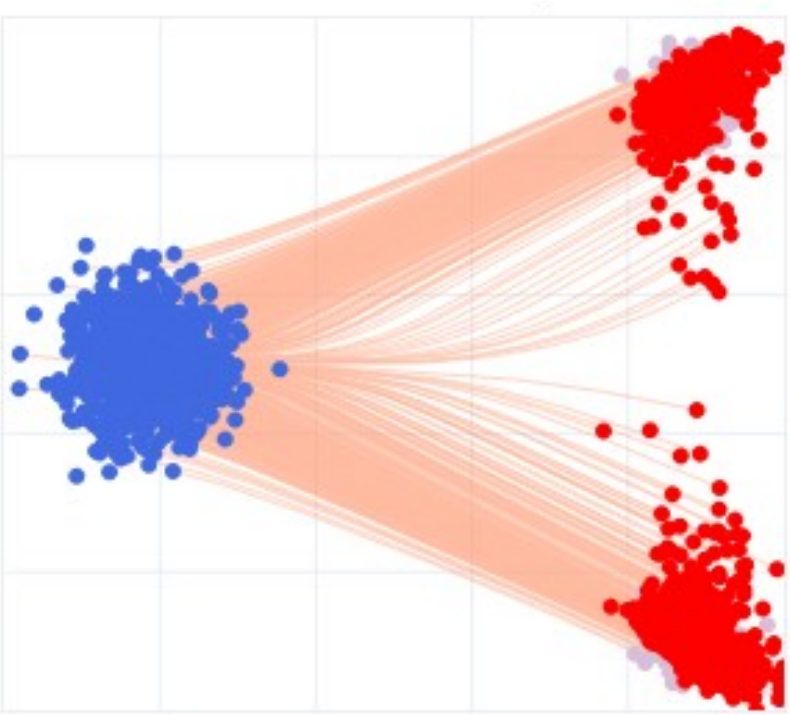
[1]Lipman, Y et al. Flowmatching for generative modeling. In ICLR. OpenReview.net, 2023.
[2]Liu, X et al. Flow straight and fast: Learning to generate and transfer data with rectified flow. In ICLR, 2023.

Rectified flow known (prior)

- Transform points z1 of Gaussian to points z0 of data distribution.
- Learns a flow defined by the ODE

$$\mathrm{d}Z_t = f(Z_t, t)\mathrm{d}t$$

, where *f* is a neural network.



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Learning rectified flow

- Learn flow by regressing direction (z0 - z1) between random samples of data, noise pairs (z0, z1), from convex combinations of those pairs

z_0 $\quad$ $(1-t)z_1 + tz_0$ $\quad$ z_1

$$\mathcal{L} = \mathbb{E}_t[\| (z_0 - z_1) - f_\theta((1-t)z_1 + tz_0, t)\|_2]$$

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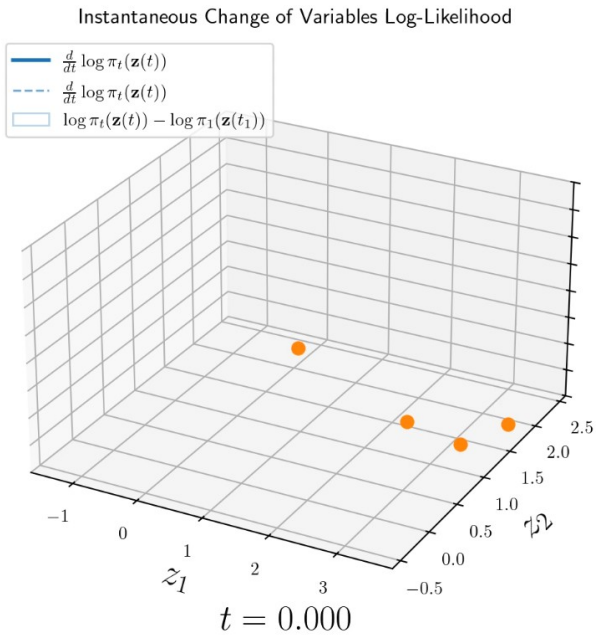
Instantaneous change of variables[1]

- Calculate IC by selecting a data point and numerically integrating from $t=t_0$ to $t=t_1$

$$\int_t^{t_1} \underbrace{\begin{bmatrix} -f(\mathbf{z}(t),t) \\ \mathrm{tr}\ \frac{\partial f}{\partial \mathbf{z}}(\mathbf{z}(t),t) \end{bmatrix}}_{\text{dynamics}} dt = \underbrace{\begin{bmatrix} \mathbf{z}_1 \\ \log \pi_t(\mathbf{z}_t) - \log \pi_1(\mathbf{z}_1) \end{bmatrix}}_{\text{solutions}}$$

- Final solution is found by adding $\log \pi_1(z_1)$



[1] Chen TQ., et al.. Neural ordinary differential equations. In NeurIPS, pages 6572–6583, 2018

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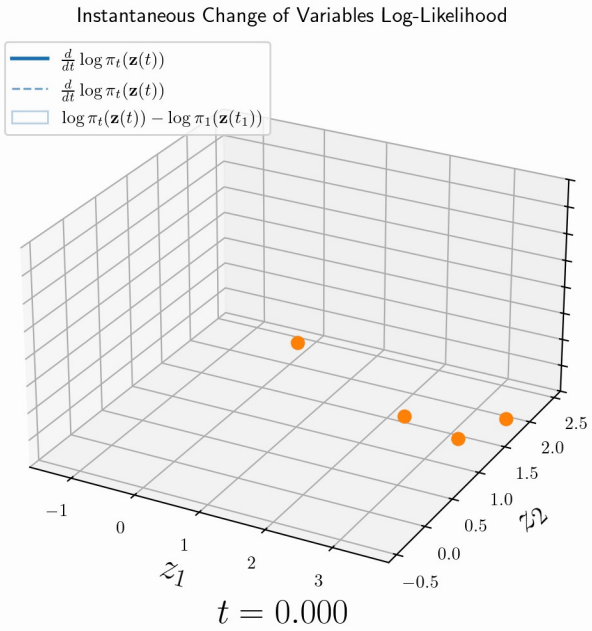

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IC estimation noised

- We can also calculate IC of "**noised**" data point by numerically integrating from $t \neq t_0$ to $t = t_1$

$$\int_t^{t_1} \underbrace{\begin{bmatrix} -f(\mathbf{z}(t), t) \\ \operatorname{tr} \ \frac{\partial f}{\partial \mathbf{z}}(\mathbf{z}(t), t) \end{bmatrix}}_{\text{dynamics}} dt = \underbrace{\begin{bmatrix} \mathbf{z}_1 \\ \log \pi_t(\mathbf{z}_t) - \log \pi_1(\mathbf{z}_1) \end{bmatrix}}_{\text{solutions}}$$

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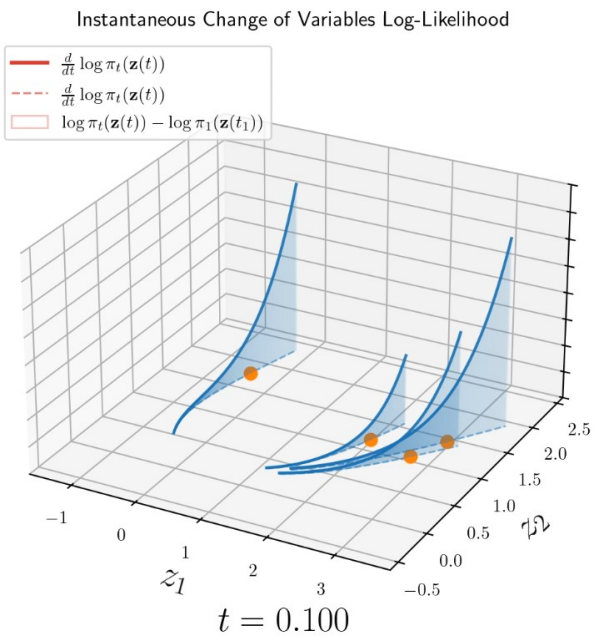

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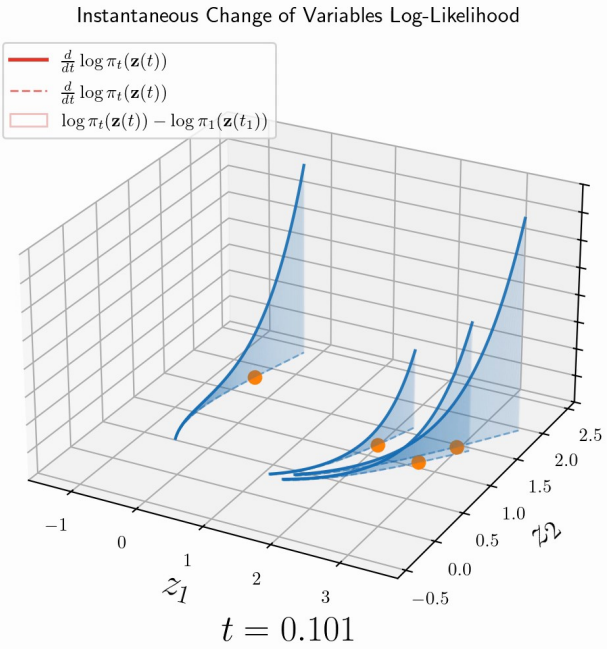


Why use noised data?

Hypothesis - Noised samples preserves most important information[1]

[1] Bjare,MR. Estimating musical surprisal from audio in autoregressive diffusion model noise spaces. In ISMIR, 2025

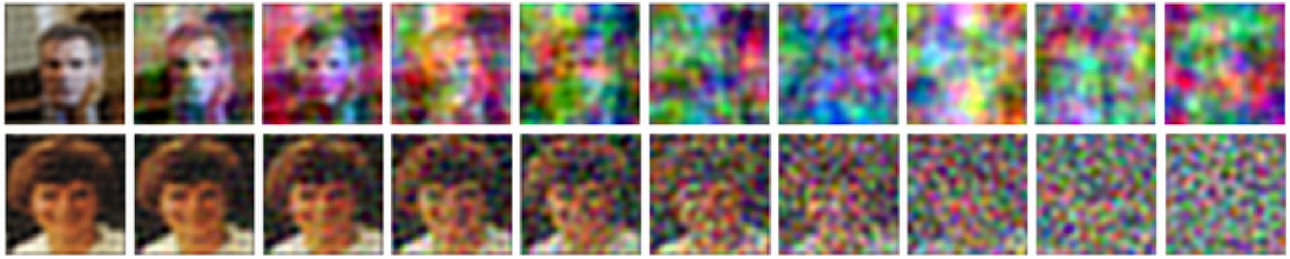
Controlling the hierachy in which Information appear

- In the pixel domain, spectral analysis reveals:
  - Low frequencies of image tend to encode objects
  - High frequencies tend to encode texture

- The order which information apper can be controlled[1]

Noise frequencies equally

Noise high freq first



[1] Falck F., A fourier space perspective on diffusion models. CoRR, abs/2505.11278, 2025.

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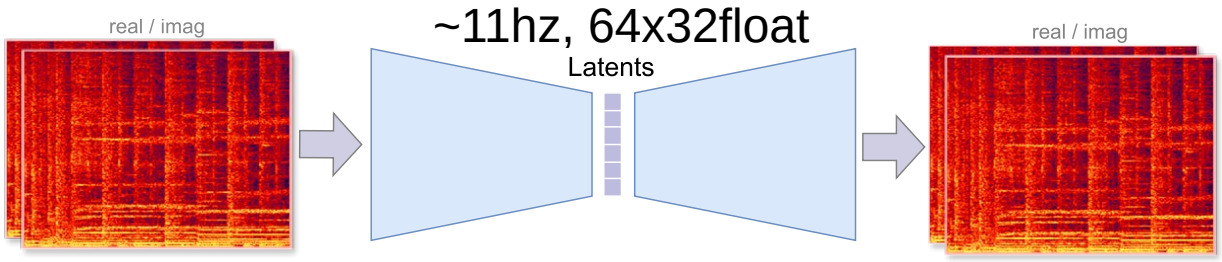
Controlling the hierachy in which Information appear

- In Bjare 2025a[1] a hierachy is assumed to exist in audio representations
- In Bjare 2025b[2] such hierachy in audio representations is *explicitely* learned.
  - we force the most important information to be encoded in noise-resistent structures.



[1] Bjare,MR. Estimating musical surprisal from audio in autoregressive diffusion model noise spaces. In ISMIR, 2025
[2] Bjare, MR, et al. Perceptually Aligning Representations of Music via Noise-Augmented Autoencoders. NeurIPS - AI for Music Workshop. 2025

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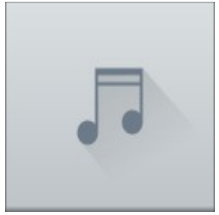
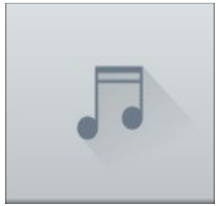
Sound example

Sound example

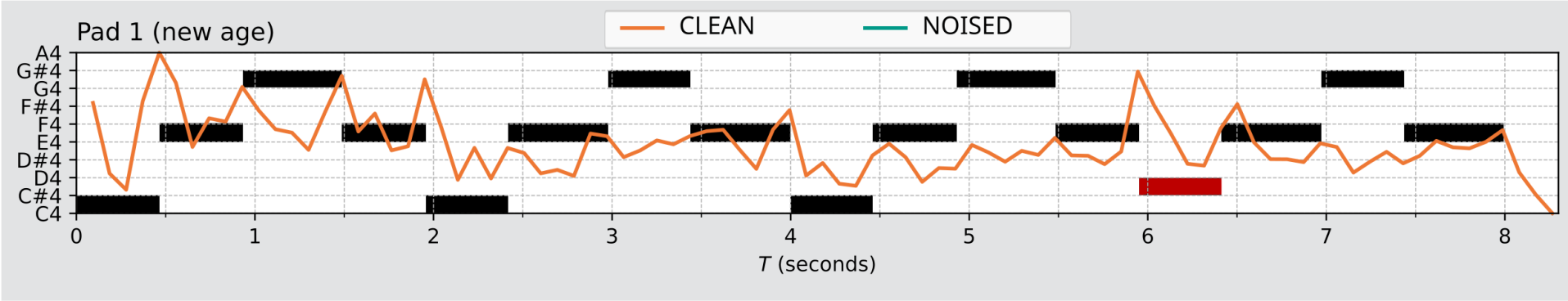


Sound example in noised space



How does our IC surprise estimations look?

Pitch surprisal: IC of sample with or without noise



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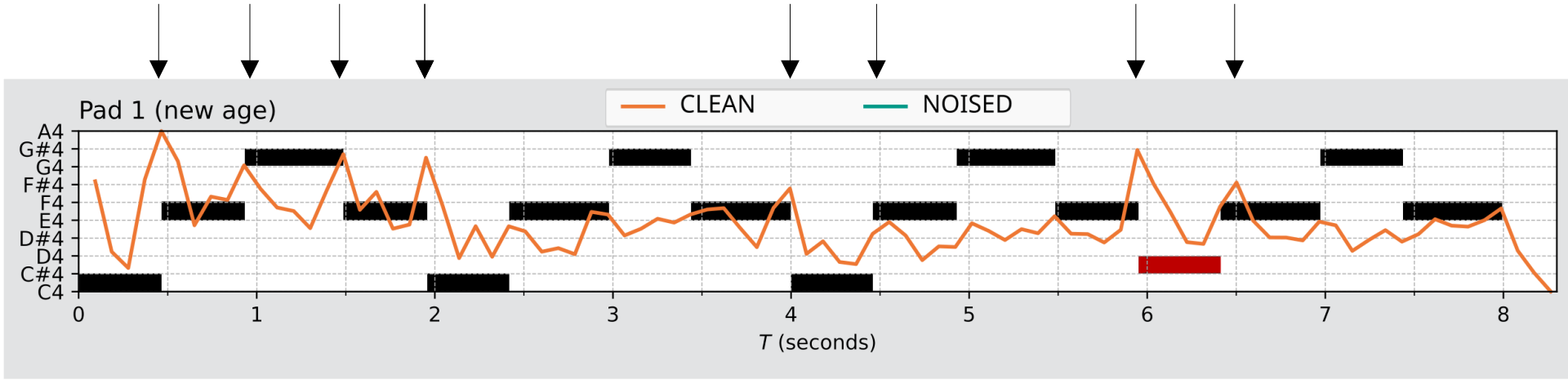
Pitch surprisal: IC of sample with or without noise



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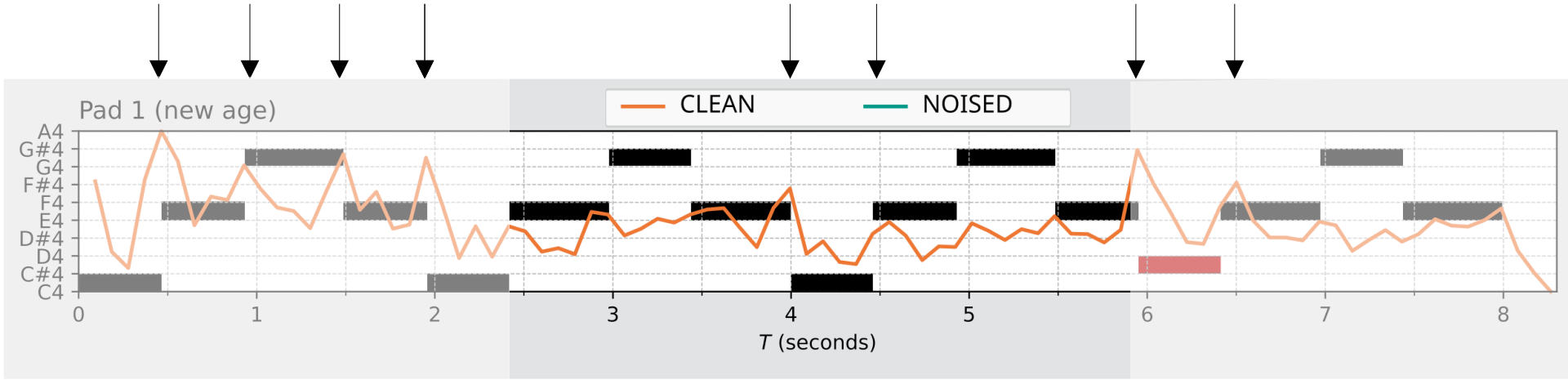

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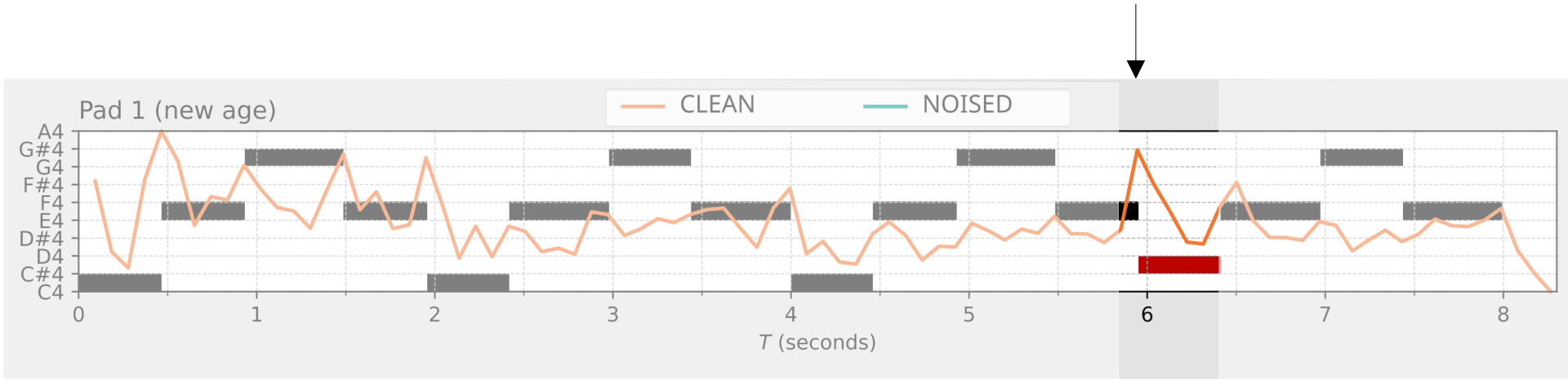

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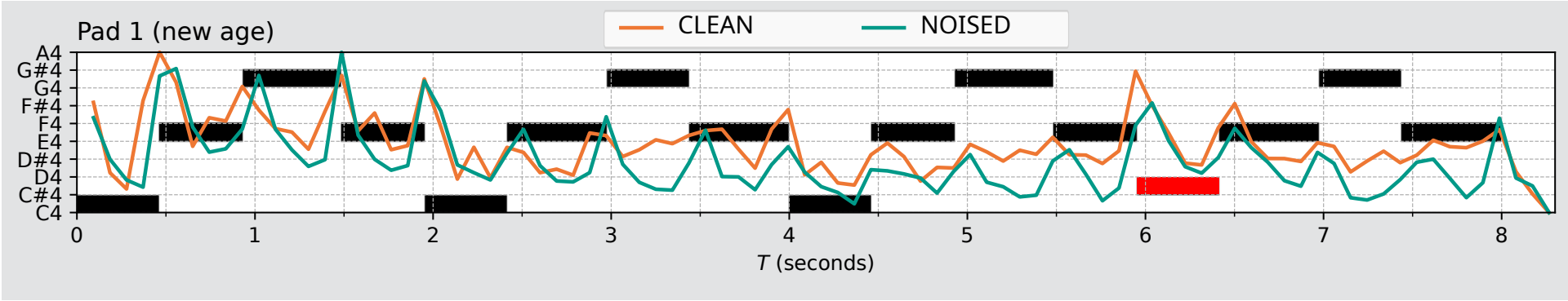

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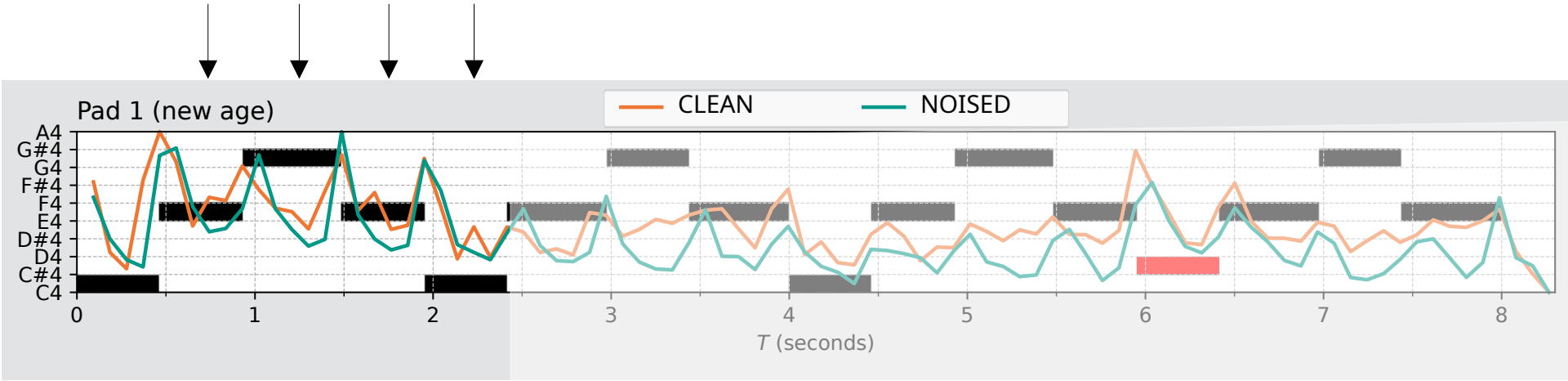

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Qualitative experiments

- Correlations of density-based IC and audio/music features[1]
- Music segments[1,2]:
  - Repetition lowers IC
  - Later segments tend to have higher IC
  - IC aligns with segment boundaries
- Correlation with IDyOM on vocal music / synthetic music[2,3]
- Neural encoding of IC (EEG responses to vocal music)[1,3]

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[1] Bjare,MR. Estimating musical surprisal in audio. In ICASSP, 2025
[1] Bjare,MR. Estimating musical surprisal from audio in autoregressive diffusion model noise spaces. In ISMIR, 2025
[2] Bjare, MR, et al. Perceptually Aligning Representations of Music via Noise-Augmented Autoencoders. NeurIPS - AI for Music Workshop. 2025

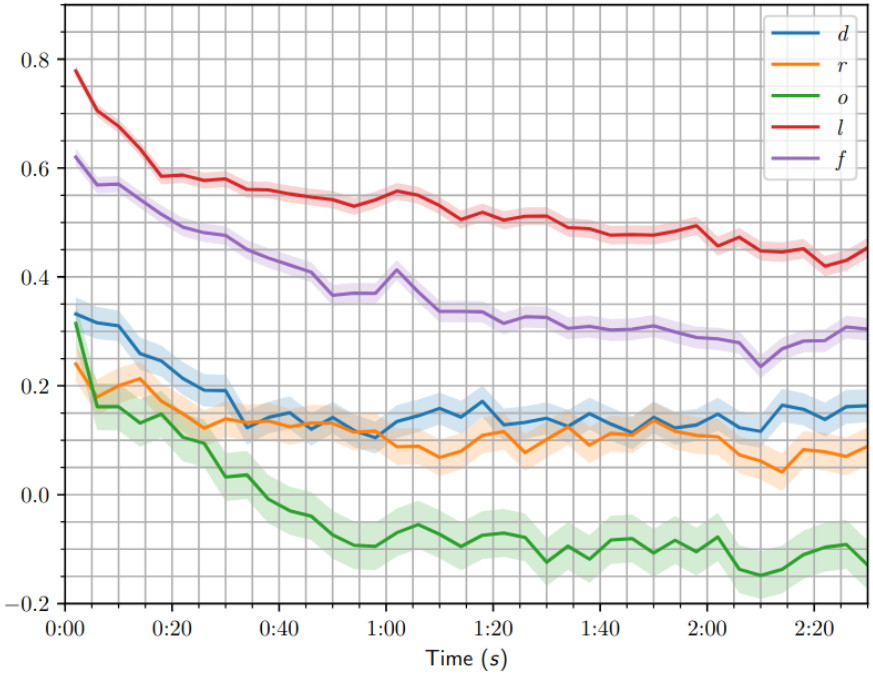
Python Package

- https://github.com/CPJKU/pa-audioic

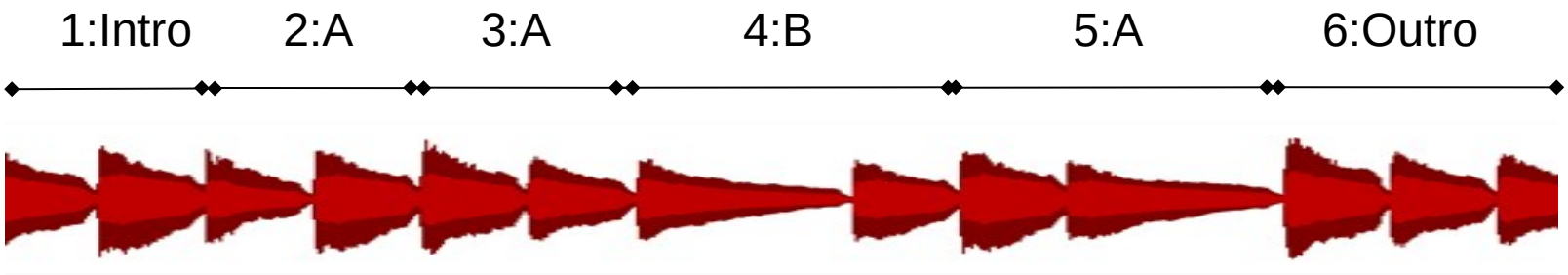
Qualitative Experiments

Correlation Studies (Bjare et al. 2025c)

- Dissonance (d)
- Rhythmic complexity (r)
- Onset density (o)
- Timbre variations (f)
- Loudness (l)



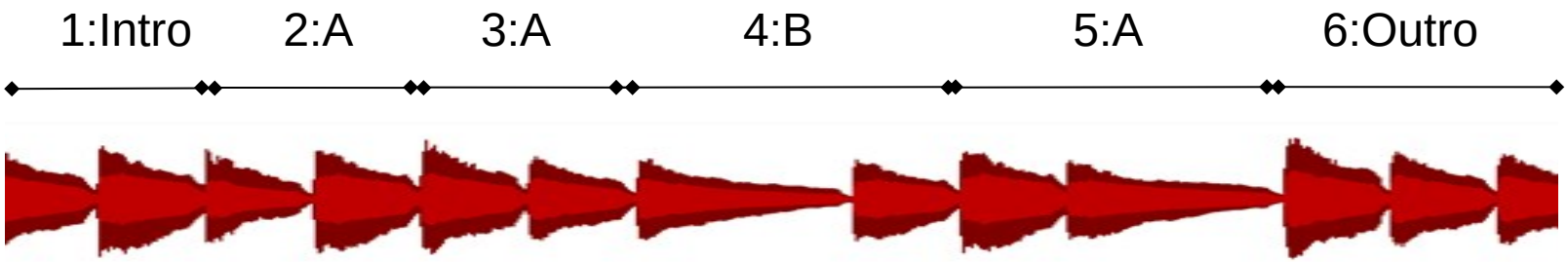
Repetition lowers IC



A **repetition** is a pair of segment indices where the corresponding segments have the same label, i.e., there are two repetitions: (2,3) and (3,5).

	Repetition μ
MU	-0.15
ME	0.03
PR	-1.30

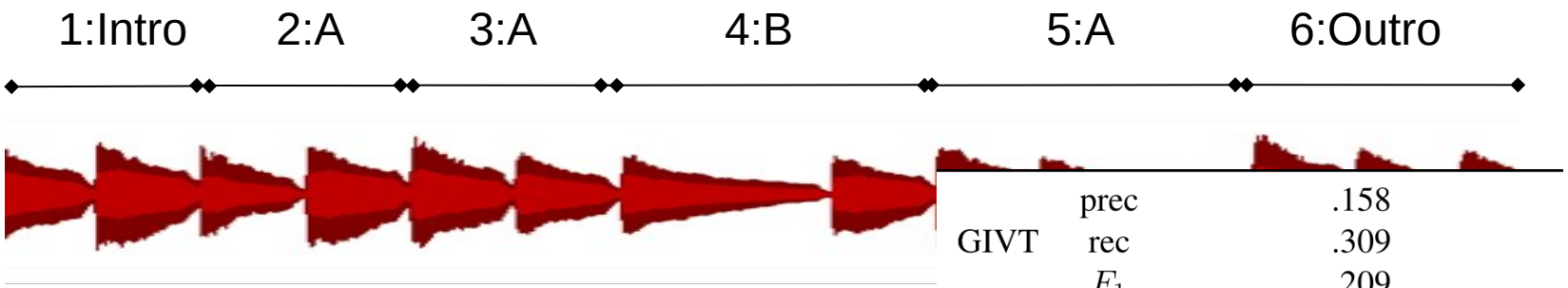
Contrasting Segments Hypothesis



We extract index pairs corresponding to subsequent sequence types **on their first occurrence**: (1,2), (2,4) and (4,6)

	Seg. Contrast	
	μ_s	μ_o
MU	2.33	-2.33
ME	1.94	-3.71
PR	3.46	-19.0

Segment boundaries alignment



- Pick peaks of the IC function
- Measure performance of segment boudrary detection by identification with IC peaks

GIVT	prec		.158		
	rec		.309		
	F_1		.209		
EDM	t	2e-3	17.6	40.0	60.0
	prec	.159	.162	.169	**.178**
	rec	.286	.311	.324	**.345**
	F_1	.204	.213	.222	**.235**
RFF	t	0.0	0.25	0.50	0.70
	prec	.159	.163	.179	**.198**
	rec	.287	.342	.380	**.416**
	F_1	.205	.221	.243	**.268**

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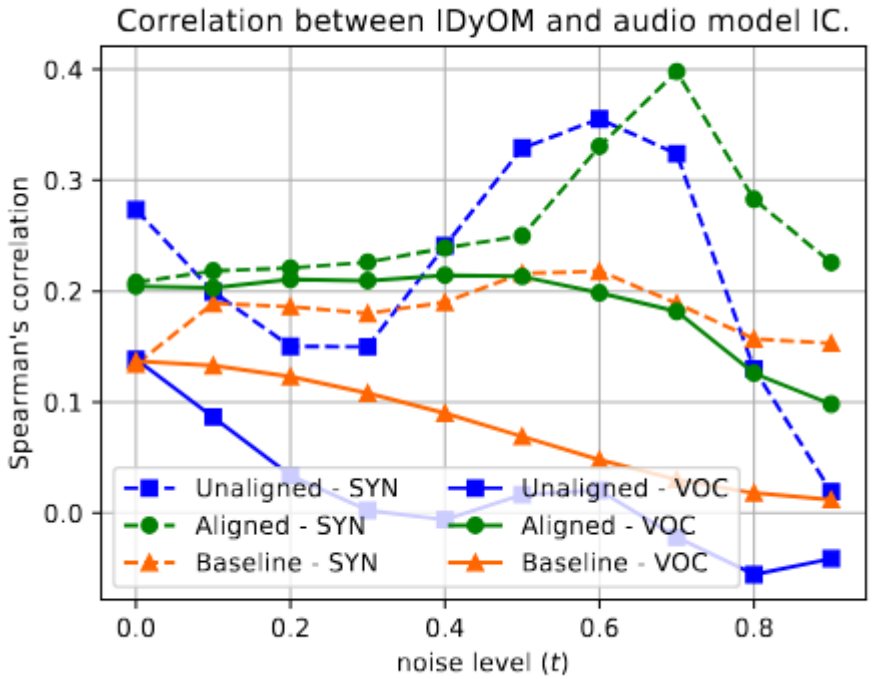
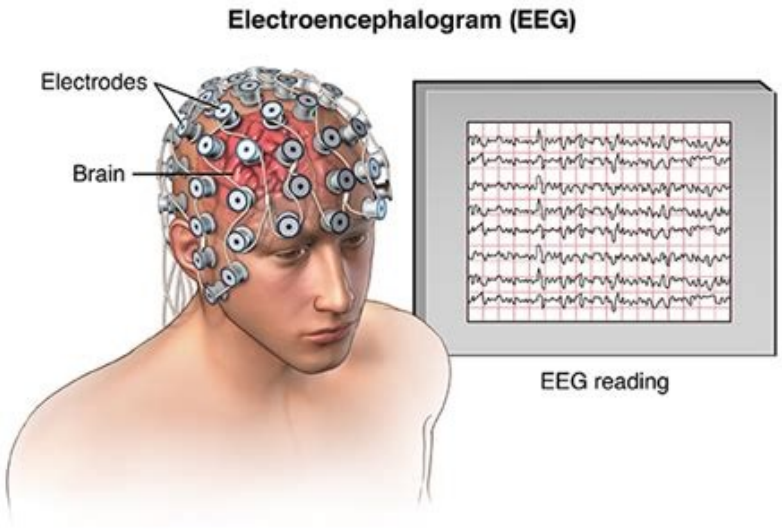
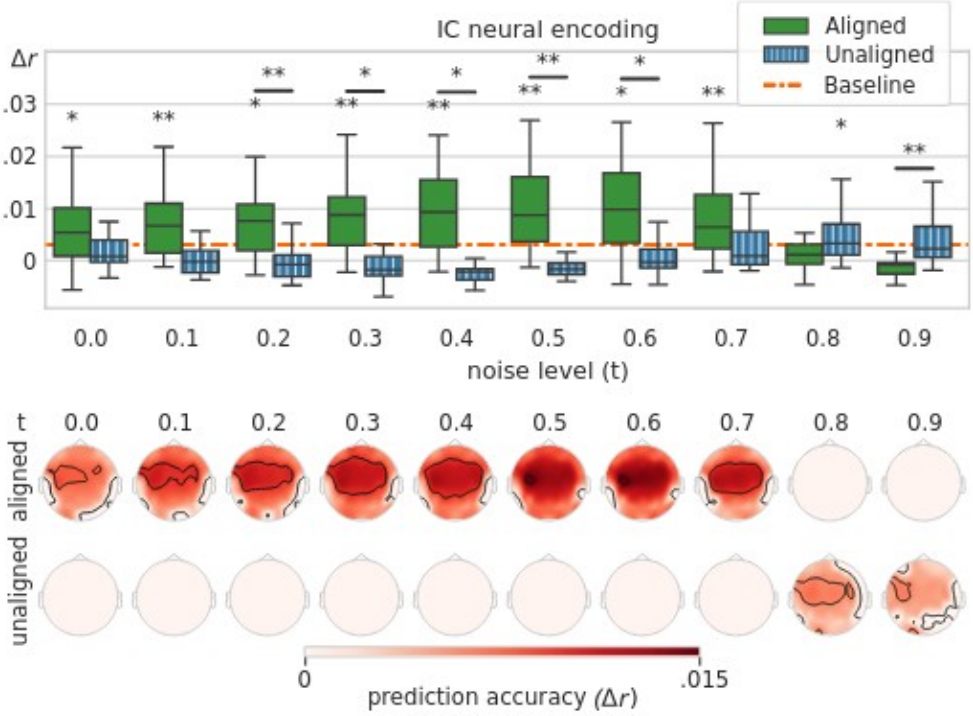
Correlation between IDyOM and audio model

Correlation between IC calculated in audio and IDyOM IC

- Recorded vocal/transcribed
- Synthesized Tunes/handwritten ABC



Prediction of EEG Responses



Prediction of EEG Responses